

Matthew H. Faust

✉ mfaust@msu.edu
🌐 mattfaust.github.io/

Positions

Aug. 2024– **Research Associate**, *Department of Mathematics, Michigan State University*,
Present Mentor: Ilya Kachkovskiy
Fall 2026 **Research Fellow**, *Simons Institute*,
Program: Spectral Theory Beyond Graphs

Education

Aug. 2018– **PhD Mathematics**, *Texas A&M University*,
Aug. 2024 Doctoral advisors: Wencai Liu and Frank Sottile
Aug. 2014– **B.S. Computer Engineering and B.S. Mathematics**,
May 2018 *University of Illinois at Urbana-Champaign*

Publications

1. *A counterexample to Fermi isospectral rigidity for two dimensional discrete periodic Schrödinger operators* by T. Brysiewicz, M. Faust, W. Liu, Mar 2026, *Selecta Mathematica* to appear.
2. *Likelihood Correspondence of Statistical Models*. D. Barnhill, J. Cobb, M. Faust, ISSAC 26', 2026, DOI: 10.1145/3815436.3815479
3. *Extrema of spectral band functions of two dimensional discrete periodic Schrödinger operators*. M. Faust, W. Liu, E. Luo, *Journal of Mathematical Physics*, 2025, DOI: 10.1063/5.0261018
4. *Irreducibility of the Dispersion Polynomial for Periodic Graphs*. M. Faust, J. Lopez Garcia, *SIAM Journal on Applied Algebra and Geometry*, 2025, DOI: 10.1137/23M1600256
5. *Floquet Isospectrality of the Zero Potential for Discrete Periodic Schrödinger Operators*. M. Faust, W. Liu, R. Matos, J. Plute, J. Robinson, Y. Tao, E. Tran, C. Zhuang, *J. Math. Phys.* 65 2024, DOI: DOI: 10.1063/5.0201744
6. *Critical Points of Discrete Periodic Operators*. M. Faust, F. Sottile, *J. Spectr. Theory*, 2024, DOI 10.4171/JST/503.
7. *The Surprising Accuracy of Benford's Law in Mathematics*. Z. Cai, M. Faust, A.J. Hildebrand, J. Li, Y. Zhang, *The American Mathematical Monthly*, 2020, DOI: 10.1080/00029890.2020.1690387.
◦ *Subject of a 2021 Paul R. Halmos - Lester R. Ford Award*
8. *Leading Digits of Mersenne Numbers*, Z. Cai, M. Faust, A.J. Hildebrand, J. Li, Y. Zhang, *Experimental Mathematics*, 2019, DOI: 10.1080/10586458.2018.1D551162.

Preprints

1. *The Dry Ten Martini Problem at Criticality*, D. S. Borgnia, M. Faust, September 2026, DOI: 10.48550/ARXIV.2112.06869
2. *Reducibility and Diagonal Equality Strata for Finite Jacobi Pencils*, M. Faust, August 2026, DOI:TBD
3. *Generic Irreducibility of Bloch Varieties for Periodic Graph Operators*, M. Faust, W. Liu, May 2026, DOI: 10.48550/ARXIV.2605.01212
4. *Common Real Secants to Pairs of Real Twisted Cubic Curves* by S. Aslam, M. Faust, J. D. Hauenstein, J. Lopez Garcia, B. Kagy, M. H. Regan, C. W. Wampler, A. Zhang, Mar 2026, DOI: 10.48550/ARXIV.2603.25003
5. *Algebraic Properties of the Ideal of Spectral Invariants for the Discrete Laplacian*. M. Faust, L. Friedman, G. O'Malley, R. Ramos, A. Sharma, Feb 2026, DOI: 10.48550/ARXIV.2602.08304
6. *Inverse Eigenvalue Problems, Floquet Isospectrality and the Hilbert–Chow Morphism*. J. Cobb, M. Faust, A. Kretschmer, Jan 2026, DOI: 10.48550/ARXIV.2601.09604
7. *The Critical Point Degree of a Periodic Graph*. M. Faust, J. Robinson, F. Sottile, Dec 2025, DOI: 10.48550/ARXIV.2512.01226

8. *The Spectral Edges Conjecture via Corners*. M. Faust, F. Sottile, Oct 2025, DOI: 10.48550/ARXIV.2510.10143
9. *Absence of flat bands for discrete periodic graph operators with generic potentials*. M. Faust, I. Kachkovskiy, Sep 2025, DOI: 10.48550/ARXIV.2509.01927
10. *Rare Flat Bands for Periodic Graph Operators*. M. Faust, W. Liu, Mar 2025, DOI: 10.48550/ARXIV.2503.03632
11. *LikelihoodGeometry: Macaulay2 Package*. David Barnhill, J. Cobb, M. Faust, Nov 2024, DOI: 10.48550/ARXIV.2411.11165

Awards

2021 **2021 Paul R. Halmos - Lester R. Ford Award**

Software Contributions

- NormalToricVarieties package for Macaulay2 (contributor).
- Likelihood Geometry package for Macaulay2 (with Barnhill and Cobb).

Travel Awards

2025 **SIAM Early Career Travel Award for travel to SIAM AG25**

2023 **JMM Graduate Student Travel Grant**

Long and Short Term Visits

- Spring 2027 **IPAM: Long Program**, *Numerical Algebraic Geometry and Correlated Electrons: Generalized Grassmannians, Response Functions and Excited States*, Mar. 8-June 11
- Fall 2026 **Simons Institute: Research Program**, *Spectral Theory Beyond Graphs*, Aug. 24-Dec. 11
- May 2026 **Collaborate@ICERM**, *Algebraic Geometry in Spectral Theory*, May 4-8
- Fall 2023 **IMSI: Algebraic Statistics and our Changing World**, Sept. 18-Dec.15

Conference and Seminar Organization

1. Special Session Organizer: Applied and Computational Algebraic Geometry in Mathematical Physics JMM 2027 (Jan 12-15 2027).
2. Special Session Organizer: Spectral Theory of Schrödinger Operators and Related Structures in Mathematical Physics JMM 2027 (Jan 12-15 2027).
3. Mini-symposium Organizer: Spectral Theory of Continuous and Discrete Schrödinger Operators and Related Topics SIAM TX-LA (Oct 30-Nov 1 2026).
4. Special Session Organizer: Periodic and Ergodic Schrödinger Operators, The 15th AIMS Conference (July 6-10 2026).
5. Special Session Organizer: Applied and Computational Algebraic Geometry AMS 2026 Spring South Eastern Sectional Meeting (March 27-29 2026).
6. Special Session Organizer: Spectral Theory of Discrete and Continuous Schrödinger Operators JMM 2026 (Jan 4-7 2026).
7. Seminar Organizer: Mathematical Physics and Operator Algebras Seminar Michigan State University Department of Mathematics, East Lansing, Michigan (2025 academic year).
8. Mini-symposium Organizer: Spectral Theory and Applications of Schrödinger Operators SIAM TX-LA (September 26-28 2025).
9. Mini-symposium Organizer: Algebraic Geometry in Spectral Theory SIAM AG25 (July 7-11 2025).
10. Special Session Organizer: Spectral Theory of Schrödinger Operators and Related Topics AMS 2025 Spring Eastern Sectional Meeting (April 5-6 2025).
11. Seminar Organizer: Mathematical Physics and Operator Algebras Seminar Michigan State University Department of Mathematics, East Lansing, Michigan (2024 academic year).
12. Special Session Organizer: Mathematics of Topological Insulators JMM 2025 (Jan 8-11 2025).
13. Special Session Organizer: Quasi-periodic and Disordered Systems AMS 2024 Fall Central Sectional Meeting (Sept 14-15 2024).

14. Mini-symposium Organizer: Discrete and Continuous Schrödinger Operators SIAM TX-LA (Nov 3-5 2023).
15. Organizer: Spectral Theory and Applications at Texas A&M (Oct 13-15 2023).

Reviewing and Refereeing

- Reviewer for MathSciNet Mathematical Reviews (2025 – present).
- Referee for various journals.

Outreach

- Aug 2025 – **MSU Professor Assistantship Program Mentor**, *Michigan State University*
Present
- August 2025 **Panelist**, *MSU postdoc orientation panel*
- May 2025 **Workshop: Applications of Commutative Algebra**, *Fields Institute*
 - Mentor for the spectral theory group with Professors Stephen Shipman and Frank Sottile.
- May – July 2025 **SURIEM REU**, *Michigan State University*, East Lansing, MI
 - Mentor for the Spectral Theory for Discrete Periodic Operators 2025 project.
- Mar. 2024 **Career Panelist**, *The 7th TX-LA Undergraduate Mathematics Conference*
- Jan. 2024 – **STODO REU 2024**, *Texas A&M University*
- Aug. 2024
 - REU project on the band edges of discrete periodic Schrödinger operators.
 - Assisted Professor Wencai Liu in leading the participating student.
- Fall 2022 **Directed reading program**, *Texas A&M University*
 - Led a student through selected readings in the philosophy of mathematics.
- Aug. 2022 – **STODO REU 2022**, *Texas A&M University*
- May 2023
 - REU project on Floquet isospectrality, resulting in a research paper.
 - Assisted Professor Wencai Liu and Dr. Rodrigo Matos in leading five students.
- Jun. 2022 – **Sottile TAMU REU 2022**, *Texas A&M University*
- May 2024
 - REU project, used a computational cluster to studying discrete periodic operators.
 - Led the student through *Ideals, Varieties, and Algorithms* by Cox, Little, O’Shea.
- Spring 2022 **Directed reading program**, *Texas A&M University*
 - Led a student through *Thinking About Mathematics* by Shapiro.
- Fall 2021 **Directed reading program**, *Texas A&M University*
 - Led a student reading *Topological Crystallography* by Sunada.
- May 2021 – **Sottile TAMU REU 2021**, *Texas A&M University*
- August 2021
 - Led two students; REU focused on the application of algebraic geometry to problems in mathematical physics.
- 2020 Spring **Directed reading program**, *Texas A&M University*
 - Led a student reading *Introduction to Quantum Graphs* by Berkolaiko and Kuchment.
- 2020 **Mentor for Peer Mentoring Program**
- Summer 2019 **Panelist**, *TAMU REU graduate school discussion panel*

Teaching Experience

- Spring 2026 **Instructor for Math 299 (Transitions)**, *Michigan State University*
- Fall 2024 – **Instructor for Math 133 (Calc II)**, *Michigan State University*
- Fall 2025
- Summer 2021 **Instructor for Math 142 (Business Calc)**, *Texas A&M University*
 - Five week course, virtually lectured daily through zoom.

2018-2024 **Teaching Assistant**, *Texas A&M University*

- Fall 2023 Math 613 (Graph Theory)
- Summer 2023: Math 601 (Methods of Applied Mathematics)
- Summer 2022: Math 308 (Differential Equations)
- Spring 2022: Math 416 (Modern Algebra II)
- Fall 2021: Math 147 (Calculus I for Biological Sciences)
- Spring 2021: Math 415 (Modern Algebra I)
- Fall 2020: Math 300 (Foundations of Mathematics)
- Summer 2020: Math 151 (Engineering Mathematics I)
- Spring 2020: Math 151 (Engineering Mathematics I)
- Fall 2019: Math 152 (Engineering Mathematics II)
- Spring 2019: Math 220 (Foundations of Mathematics)
- Fall 2018: Math 220 (Foundations of Mathematics)

2016-2017 **Course Assistant**, *University of Illinois*

- Fall 2017: Math 231 (Calculus II)
- Fall 2016: CS 374 (Introduction to Algorithms)
- Spring 2016: CS 374 (Introduction to Algorithms)

Programming

- Adept: Maculay2, Python, Sage.
- Familiar: Java, PyTorch, Haskell, SystemVerilog, C, C++, x86 Assembly, JavaScript, Julia.

Upcoming Presentations

- JMM AMS Special Session on The Mathematics of Quantum Many-Body Systems and Related Topics, Chicago, Illinois, January 15, 2027
- AMS Fall Eastern Sectional special session on Spectral Theory, Ergodicity, and Mathematical Physics, George Washington University, Washington, DC, October 3-4, 2026

Recent Presentations

1. The 15th AIMS Conference special session on Dynamics and Spectrum of Quasiperiodic Schrodinger Operators, Athens, Greece, July 6-10, 2026.
2. University of California Berkeley AMO/QI Seminar, Berkeley, California, April 29, 2026.
3. Ohio River Analysis Meeting 15 (ORAM) at University of Kentucky, Lexington, Kentucky, March 21, 2026
4. JMM AMS Special Session on Numerical Algebraic Geometry and its Applications, Washington D.C, January 6, 2026
5. Western University Geometry and Combinatorics Seminar, London, Ontario, Canada, December 8, 2025
6. SIAM TX-LA: Spectral Theory and Applications of Schrödinger Operators Mini-symposium, University of Austin, Austin, TX, September 28, 2025
7. Texas A&M Geometry Seminar, College Station, TX, September 22, 2025
8. QMath 16, Munich, Germany, September 4, 2025
9. SIAM AG25: Algebraic Geometry in Spectral Theory Mini-symposium, University of Wisconsin, Madison, Wisconsin, July 11, 2025
10. Great Lakes Math Physics (GLaMP), University of Kentucky, Lexington, Kentucky, June 6, 2025
11. Workshop on the Applications of Commutative Algebra, Fields Institute, Toronto, Ontario, Canada May 27, 2025
12. Analysis and PDE Seminar, University of Kentucky, Zoom, March 4, 2025
13. MPOA Seminar, Michigan State University, East Lansing, Michigan, September 17, 2024
14. AMS Fall Central Section Meeting Special Session on Periodicity in Quantum Systems, UTSA, San Antonio, Texas, September 14, 2024
15. Early Career Workshop in Mathematical Physics, Texas A&M University, College Station, Texas, September 13, 2024
16. Effective Methods in Algebraic Geometry (MEGA), Max Planck Institute, Leipzig, Germany, August

1, 2024

17. GLaMP, Michigan State University, East Lansing, Michigan, June 8, 2024
18. Workshop on Computational and Applied Enumerative Geometry, Fields Institute, Toronto, Ontario, Canada June 6, 2024
19. Texas A&M Algebra and Combinatorics Seminar, College Station, Texas, April 19, 2024
20. AMS Eastern Spring Sectional Session on Spectral Theory and Quantum Systems, Howard University, Washington D.C, April 6, 2024
21. CombinaTexas, Texas A&M University, College Station, Texas, March 23, 2024
22. JMM AMS Special Session on Spectral Methods in Quantum Systems: San Francisco, CA, Jan 6, 2024
23. Algebraic Statistics and Our Changing World: Long Program, IMSI Chicago, Illinois, Dec 4, 2023
24. SIAM TX-LA: (Complex) analysis in differential equations, Lafayette, Louisiana, November 4, 2023
25. SIAM TX-LA: Applications of combinatorial and computational algebraic geometry, Lafayette, Louisiana, November 4, 2023
26. Algebraic Statistics and Our Changing World: Long Program, IMSI Chicago, Illinois, October 18, 2023
27. Algebraic Statistics and Our Changing World: Apprenticeship week, IMSI Chicago, Illinois, October 4, 2023
28. GLaMP, Oberlin Ohio, June 10, 2023
29. Copenhagen-Jerusalem Combinatorics Seminar, Zoom, March 30, 2023
30. AMS Spring Southeastern Section Meeting, Special Session on Quasi-periodic Schrödinger operators and quantum graphs, Atlanta, GA, March 19, 2023
31. JMM AMS Special Session on Applied Enumerative Geometry, Boston, MA, January 4, 2023
32. SIAM-TXLA 2022, Houston, Texas, November 5, 2022
33. Louisiana State University Applied Analysis Seminar, Baton Rouge, Louisiana, October 11, 2022
34. Texas A&M Algebra and Combinatorics Seminar, College Station, Texas, October 7, 2022
35. QMath15, Davis, California, September 14, 2022
36. MEGA, Krakow, Poland, June 23, 2022
37. GLaMP, Michigan State University, East Lansing, Michigan, June 11, 2022
38. JMM AMS Special Session on Structured Polynomial Systems In Mathematics and Its Applications, Zoom, April 6, 2022
39. AMS Spring Sectional, Zoom, March 26, 2022
40. Conference on spectral theory of ergodic quantum systems, Zoom, March 12, 2022
41. Texas A&M MPHA seminar, College Station, Texas, March 11, 2022
42. SIAM TX-LA, South Padre Island, Texas, Nov. 7, 2021
43. Texas A&M MPHA seminar, College Station, Texas, April 30, 2021